Stack Overview

Function	*Name*	*Dose*	*Frequency*	*Pattern*
Metabolic	**Retatrutide**	SC 1.7 mg	Su, We	Continuous
	Metformin	PO 500 mg	Daily	Continuous
GH-axis	**Tesamorelin**	SC 1 mg	AM (fasted) PM (fasted)	12 weeks on / 6 weeks off
Tissue repair	**GHK-Cu**	SC 1-2 mg	Daily	Alternating vials
	GLOW (GHK-Cu, Tβ4, BPC-157)	SC 1.2-2.4 mg	Daily	
	ARA-290	SC 4 mg, 2 mg, 2 mg	Tu, Th, Sa	Continuous
Anti-inflammatory-	**KPV**	SC 750 mcg	Daily	Continuous
	Omega-3	PO EPA 660 mg; DHA 440 mg	Daily, food	Continuous
Immune modulation	**Thymosin α1**	SC 1.0 mg	Su, We	Continuous
	Thymulin	SC 1.0 mg	M,T,Tr, Fr,Sa	Continuous
	Zinc Picolinate	PO 30 mg elemental zinc	3× weekly	Continuous
Longevity	**Spermidine 3HCl**	PO 10 mg	Daily (evenings)	Continuous
	Tadalafil	PO 5 mg	Daily	Continuous
	Multivitamin	PO 1 capsule	Daily, food	Continuous
Mineral support	**Iron Bisglycinate**	PO 25 mg elemental iron	Daily, w food	Continuous
Neurotrophic	**Semax + Pinealon**	IN 900 mcg + 900 mcg	Daily	Occasionally
Sleep support	**Epitalon + DSIP**	IN 200 + 200 mcg	Daily, 1 hour before	Continuous
	L-Glycine	PO 3000 mg	bedtime	Continuous
Cellular Energy and Brain Support	**NMN**	PO 1000 mg	Daily	Continuous
	Acetyl-L-Carnitine (ALCAR)	PO 1000 mg	Daily	Continuous
	CoQ10	PO 200 mg	Daily, food	Continuous
	Creatine Monohydrate	PO 3000 mg	Daily	Continuous
	PQQ	PO 20 mg	Daily	Continuous
	Apigenin	PO 200 mg	Daily, food	Continuous
	Beta-alanine	PO 3000 mg	Daily	Continuous
Prostate support	**Beta-sitosterol**	PO 250 mg	Daily, food	Continuous
Hair support	**Minoxidil**	PO 1.25 mg	Daily	Continuous

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These peptides are used annually or infrequently.

Function	*Name*	*Dose*	*Frequency*	*Pattern*
Longevity	**FOXO4-DRI**	SC 25 mg	Q48H ×3	Annually
	Epitalon + Pinealon	SC 5 mg; 0.5 mg	Daily for 10 days	Annually
Prostate support	**Prostamax (KEDP)**	SC 0.5 mg	Daily for 10 days	Annually

Before You Stack

Peptides and supplements are multipliers of your fundamentals: good sleep, good nutrition, and exercise. Get your fundamentals right, then consider your goals and how supplements, peptides, etc. might help.

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations, and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

Evidence classifications used in this document:

- *Claims supported by good human evidence are simply stated without qualification.*
- *Claims supported by good animal evidence, when not contradicted by human findings, are described as likely.*
- *Claims supported by in vitro or mechanistic evidence, when not contradicted by higher-level findings, are described as plausible.*
- *Claims lacking these forms of support are omitted or, if there is strong community support, identified as speculative.*

A lack of human evidence may simply reflect that appropriate trials and experiments have not been performed because of cost, limited commercial incentives, weak intellectual-property protection, or the absence of a disease-specific regulatory endpoint. It should not automatically be interpreted as evidence against the biological mechanism or efficacy.

Lifestyle

Nutrition: I am 100% vegan, focused on a whole food diet, and limiting (but not fully eliminating) processed food.

Intermittent Fasting: I fast five to seven days a week from 7 PM until 11 AM.

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Sleep: I am focusing on sleep improvements. My bedroom is dark; the bed is temperature controlled for cool sleep. My wake time is the same each day. My average sleep duration is 8 hours at least 5 days/week, but my sleep still needs improvement.

Exercise: My resistance training schedule is two to three times each week. The training is a complete upper and lower body routine with each movement focusing on maximum intensity with 8–20 reps.

Stack Goals

My overarching goal is **health support**. The individual goals supporting this are **metabolic stability**, **body composition**, **recovery**, **immune support**, **aging support, and healthspan**. As this stack pursues these specific goals, it **never intentionally wavers from making the overall healthiest choice**.

Functional Benefits

- Retatrutide reduces body weight in people with obesity and improves glycemic control in people with type 2 diabetes. It likely increases energy expenditure. Complementary body-composition effects from combined GH-axis stimulation are plausible. [4,5,103]
- GH-axis stimulation plausibly supports recovery, tissue maintenance, and anabolic signaling. Community reports describe improved sleep quality with GH-stimulating peptide combinations.
- Likely support for tissue healing and collagen production in skin, tendons, ligaments, and other soft tissues. BPC-157 likely supports restoration of tissue perfusion under ischemic conditions. [12–16,104]
- Likely support for reduced intestinal inflammation, epithelial repair, and mucosal integrity. KPV likely reduces inflammatory-cell recruitment beyond the intestine, with plausible benefits for digestive comfort. [17,105]
- Thymulin likely helps regulate age-associated inflammation. Combined immune support from thymulin, thymosin α1, and the other stack components is plausible through modulation of innate and adaptive immune pathways. [20–24]
- PQQ improved memory test scores in human trials using approximately 20 mg/day for 12 weeks. Semax likely supports neurotrophic signaling. The combined stack plausibly supports focus, neuroplasticity, and resilience to stress. [25,26,76,77]
- Pinealon likely supports neuronal protection against oxidative stress, with plausible benefits for cognitive resilience. [33,34,101,102]

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- NMN likely supports mitochondrial function. The combined stack plausibly supports cellular energy and recovery through complementary effects on NAD^+ metabolism, mitochondrial pathways, autophagy, and mitophagy. [44,45,62,74]
- Creatine monohydrate supports repeated high-intensity exercise. Beta-alanine increases muscle carnosine and buffering capacity; performance benefits are best supported for efforts lasting approximately 1–4 minutes after sustained supplementation, commonly at 4–6 g/day. [51,56]
- FOXO4-DRI likely supports selective clearance of senescent cells and reduced senescence-associated inflammatory signaling, with plausible benefits for tissue maintenance. FOXO4-DRI likely supports clearance of senescent vascular endothelial cells and improved vascular function. [37] My FOXO4-DRI diary documents marked fading of age spots and flattening and smoothing of two longstanding raised skin lesions following the course. [36,37; personal diary]
- Epitalon likely supports restoration of age-disrupted melatonin rhythms, with plausible support for telomere maintenance through effects on telomerase activity. [28–30]
- Tadalafil and beta-sitosterol improve lower urinary tract symptoms in men with benign prostatic hyperplasia. Beta-sitosterol also improves urinary flow measures. [38,39]

Metabolic Benefits

The stack is intended to support metabolic health through complementary effects on appetite, glucose regulation, and cellular energy metabolism. Retatrutide suppresses appetite, reduces body weight, and can improve glycemic control. [1,2,4,5] Its glucagon-receptor activity likely contributes to increased energy expenditure, with plausible support for fat oxidation. [103] Retatrutide substantially reduces body weight and liver fat and can normalize liver fat in many people with metabolic dysfunction-associated steatotic liver disease. Metformin improves hepatic insulin sensitivity and reduces gluconeogenesis. Creatine monohydrate supports repeated high-intensity exercise. Beta-alanine increases muscle carnosine and buffering capacity; performance benefits are best supported for efforts lasting approximately 1–4 minutes after sustained supplementation, commonly at 4–6 g/day. [51,56] NMN and ALCAR plausibly support cellular energy metabolism. Together, these effects plausibly support metabolic flexibility. PQQ likely supports mitochondrial biogenesis, with plausible support for redox regulation, cellular energy function, and metabolic flexibility. [108]Together, these agents plausibly provide additive reductions in caloric intake, improvements in glycemic control, and increased energy expenditure, favoring body-composition improvement.

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GH Benefits

Tesamorelin before sleep and Tesamorelin upon waking, both 12 weeks on and 6 weeks off likely increases pulsatile GH secretion and circulating IGF-1 above baseline. The regimen plausibly shifts GH/IGF-1 markers toward values more typical of middle-aged adults, with the actual response assessed through laboratory monitoring. [7,9–11]

Regenerative Benefits

This stack combines peptides whose proposed mechanisms act at complementary points in the repair cascade and could plausibly provide additive signaling for matrix remodeling, inflammation control, and tissue repair. GHK-Cu likely supports collagen production and tissue repair, with plausible modulation of MMP/TIMP balance contributing to extracellular-matrix remodeling; TB-500 (TB4) likely supports tissue repair through effects on cell migration, angiogenesis, and extracellular-matrix remodeling. *(In this document, "TB-500 (TB4)" refers to full-length, 43-amino-acid thymosin β4, rather than the shorter LKKTETQ fragment.)* BPC-157 likely supports tendon, ligament, and gut mucosal healing and restoration of tissue perfusion under ischemic conditions, with plausible effects on local growth-factor signaling, epithelial restitution, and microvascular stability [104]; and KPV likely reduces intestinal inflammation, with plausible inhibition of NF-κB-driven cytokine release and support for epithelial-barrier recovery. [17]

ARA-290 improved neuropathic symptoms and increased corneal nerve-fiber density in human trials of sarcoidosis-associated small-fiber neuropathy. It likely supports broader tissue protection, microvascular preservation, and cutaneous wound healing. [18,19,106]

Tesamorelin likely increase pulsatile GH secretion and circulating IGF-1 above baseline, with plausible support for recovery, tissue maintenance, and anabolic repair processes. [7,9–11]

Prostamax (KEDP) is speculative. Some practitioners use it with the aim of supporting local prostate tissue regulation and microcirculation, reducing inflammation, and promoting tissue homeostasis.

Immune Benefits

Thymosin α1 likely supports dendritic-cell activation and Th1-mediated antifungal immunity. Repeated dosing plausibly supports NK-cell activity, broader T-cell responses, and interferon-associated pathways without requiring continuous circulating peptide. [107]

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Thymulin is a zinc-dependent thymic regulatory hormone whose circulating biological activity declines substantially with age. Beyond its established relationship to thymic and T-cell regulation, recent experiments identify thymulin as a regulator of age-associated inflammatory myeloid cells. Research in aged human immune cells and aged mice shows thymulin suppresses IL-1α, IL-1β, IL-6, and TNF-α production through inhibition of canonical NF-κB signaling while improving T-cell function. Effects were substantially greater in aged than young immune systems, supporting a model in which thymulin helps normalize age-related immune dysregulation rather than simply stimulating or suppressing immunity. (Zinc supplementation is included to maintain adequate zinc status and to plausibly support the conversion of zinc-free thymulin into active Zn-thymulin.)

Thymosin α1 and thymulin may provide complementary rather than redundant immune modulation. Thymosin α1 plausibly supports coordination of innate and adaptive immune responses. [20,21] Thymulin likely suppresses age-associated inflammatory myeloid-cell activity through regulation of NF-κB signaling. [22] The combination therefore plausibly addresses both immunosenescence and inflammaging.

Sleep Benefits

Epitalon, DSIP, and L-Glycine, combined with Tesamorelin, target distinct but potentially overlapping aspects of sleep regulation and regeneration. Together, these agents could plausibly support a sleep environment by involving circadian timing, sleep-related neuroendocrine regulation, thermoregulatory cooling, nocturnal GH signaling, and overnight recovery.

Tadalafil improves BPH-associated lower-urinary-tract symptoms and may reduce nocturia-related sleep disruption. Beta-sitosterol improves urinary symptoms in men with benign prostatic hyperplasia, while Prostamax remains speculative. [38]

Healthspan Benefits

A ***conserved pathway of aging*** is a biological process that shows up across many species, changes predictably with age, and when experimentally modified can speed up or slow down aging. These pathways represent the deep, evolutionarily preserved mechanisms that drive cellular decline, tissue dysfunction, and loss of resilience over time.

If a therapy targets a conserved pathway of aging and if interventions on that pathway reliably slow aging in model organisms, then modifying that same pathway in humans is plausibly healthspan-promoting, even if direct human longevity data are not yet available.

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The stack is organized below against the twelve hallmarks of aging described by López-Otín and colleagues. These hallmarks provide a framework for assessing mechanistic relevance; targeting a hallmark does not itself establish a human healthspan benefit. [48]

Genomic instability — accumulation of DNA damage and impaired maintenance of genome integrity.

Telomere attrition — shortening or dysfunction of chromosome-end protective structures.

Epigenetic alterations — changes in chromatin and gene regulation that disrupt cellular function.

Loss of proteostasis — impaired protein folding, quality control, and disposal.

Disabled macroautophagy — impaired cellular recycling of damaged proteins and organelles.

Deregulated nutrient sensing — altered signaling governing nutrient availability, metabolism, growth, and maintenance.

Mitochondrial dysfunction — impaired mitochondrial energy production and cellular signaling.

Cellular senescence — accumulation of persistently arrested cells with potentially harmful secretory activity.

Stem cell exhaustion — declining stem-cell function and regenerative capacity.

Altered intercellular communication — disrupted signaling between cells, tissues, and organs.

Chronic inflammation — persistent inflammatory activity that contributes to tissue dysfunction.

Dysbiosis — changes in microbial communities and their interactions with the host.

This table maps the stack to the twelve hallmarks of aging. Evidence scopes describe the specific mechanisms or outcomes studied, rather than demonstrated slowing of human aging. Components may influence several hallmarks, and their combined effects remain plausible.

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Hallmark	Relevant stack components	Evidence scope
Genomic instability	NMN; sleep and circadian practices	NMN likely supports NAD^+-dependent DNA repair; sleep and circadian practices provide plausible indirect support. [63]
Telomere attrition	Epitalon	Plausible support for telomerase activity and telomere maintenance, based on cell studies. [28,29]
Epigenetic alterations	Epitalon; Pinealon; NMN; apigenin; resistance training; sleep and circadian practices	Plausible influence on chromatin, gene regulation, and cellular metabolic signaling. Pinealon plausibly influences neuronal gene regulation and cell-survival signaling. [101,102]
Loss of proteostasis	Spermidine; time-restricted feeding; resistance training	Plausible support for protein quality control through cellular maintenance and recycling pathways. [44,45,48]
Disabled macroautophagy	Spermidine; time-restricted feeding; apigenin; metformin	Spermidine likely supports autophagy; a human pilot found increased B-cell autophagic flux in selected older adults. Apigenin likely supports restoration of stress-suppressed autophagy. Other contributions remain plausible. [44–46,72,73]
Deregulated nutrient sensing	Metformin; Retatrutide; time-restricted feeding; Tesamorelin	Human evidence supports individual metabolic and endocrine effects. Their combined influence on aging-related nutrient sensing is plausible. [1–11,49]
Mitochondrial dysfunction	NMN; ALCAR; CoQ10; PQQ; creatine monohydrate; exercise	NMN likely supports mitochondrial function, and PQQ likely supports mitochondrial biogenesis. The other components contribute to energy-related pathways, with plausible combined mitochondrial support. [62,65,68,74,108]
Cellular senescence	FOXO4-DRI; spermidine; thymosin α1; thymulin	FOXO4-DRI likely supports senescent-cell clearance. Spermidine has evidence relevant to immune-cell senescence;

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		complementary immune support remains plausible. [22,36,37,46]
Stem cell exhaustion	TB-500 (TB4); GHK-Cu; BPC-157; ARA-290; Tesamorelin; exercise	Likely tissue-repair effects from selected peptides provide a rationale for regenerative support; preservation or restoration of stem-cell reserves remains plausible. [12–19,90,91]
Altered intercellular communication	Semax; Epitalon; DSIP; glycine; Tesamorelin; thymosin α1; thymulin; tadalafil	These components influence neuronal, circadian, endocrine, immune, or vascular signaling. Their combined contribution to healthier age-related communication is plausible.
Chronic inflammation	Thymulin; KPV; ARA-290; EPA/DHA; apigenin; metformin	Thymulin likely reduces age-associated myeloid inflammation; KPV likely reduces intestinal inflammation and inflammatory-cell recruitment beyond the intestine. Complementary systemic effects remain plausible. [17,22,69,72,105]
Dysbiosis	Whole-food vegan diet; metformin; KPV; BPC-157	Metformin alters the gut microbiome in human type 2 diabetes studies. Diet provides microbial substrates; KPV and BPC-157 offer likely intestinal tissue support, with plausible indirect microbiome benefits. [17,96]

Several components serve goals that are not direct hallmark interventions: beta-alanine supports muscle buffering; beta-sitosterol and tadalafil address urinary symptoms; minoxidil supports hair growth; and Prostamax is used for practitioner-reported prostate support. Iron, zinc, and the multivitamin support nutritional adequacy. Pinealon likely supports neuronal protection against oxidative stress, with plausible benefits for cognitive resilience. [33,34,101,102]

Stack Design: Synergies

- **Retatrutide + Tesamorelin:** This combination plausibly supports complementary reductions in liver fat and visceral abdominal fat. Retatrutide's glucagon-receptor activity likely contributes to increased energy expenditure, with plausible support for fat oxidation. [3,9–11,103]

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- **GHK-Cu + TB-500 (TB4) + BPC-157 + KPV + ARA-290:** These regenerative peptides target complementary pathways involving collagen synthesis, actin remodeling, angiogenesis, epithelial repair, microvascular protection, tissue-protective signaling, and anti-inflammatory signaling, which could plausibly provide additive repair signaling.
- **Epitalon, DSIP, and L-Glycine:** This combination could plausibly support sleep through complementary and potentially overlapping effects on circadian signaling, sleep-related neuroendocrine pathways, thermoregulatory cooling, and glycine-mediated neurotransmission.
- **NMN + ALCAR + CoQ10 + Creatine + Beta-alanine + Apigenin + PQQ:** These mitochondrial and energy supplements have complementary mechanisms that plausibly support cellular energy capacity by supporting NAD+ pools, electron-transport-chain function, antioxidant defenses, fatty-acid metabolism, rapid ATP regeneration, PQQ-driven mitochondrial-biogenesis signaling, and intracellular buffering.
- **Semax + Pinealon + NMN/ALCAR:** Semax's likely neurotrophic effects and Pinealon's likely neuronal protection against oxidative stress plausibly complement the metabolic support provided by NMN and ALCAR, with the intended goals of supporting cognitive performance and neuronal resilience. [25,26,33,34,65,67,101,102]
- **Thymosin α1 + Thymulin:** Thymosin α1 likely supports dendritic-cell activation and Th1-mediated antifungal immunity, with plausible contributions from NK-cell cytotoxicity and broader interferon-associated pathways, while Thymulin contributes thymic and T-cell regulatory signaling. [107] Because these effects are distinct but potentially overlapping, the combination could plausibly improve coordination between immune activation and inflammatory regulation.
- **Tadalafil + Beta-sitosterol + Prostamax**: Daily tadalafil improves lower-urinary-tract symptoms associated with BPH. Beta-sitosterol improves urinary symptoms in men with benign prostatic hyperplasia, while Prostamax remains a speculative organ-specific intervention. [38] Together, they form a plausible prostate-support strategy involving symptom control, cGMP-mediated smooth-muscle effects, and possible local microcirculatory support.
- **ARA-290 + tadalafil + omega-3:** These agents have mechanistically complementary effects that could plausibly support microvascular and endothelial function.
- **FOXO4-DRI + Spermidine + Intermittent fasting + NMN + Resistance training:** FOXO4-DRI likely supports selective clearance of senescent cells. Spermidine likely supports autophagy, and apigenin likely supports restoration of stress-suppressed autophagy. Intermittent fasting, NMN, and resistance training plausibly contribute to cellular maintenance and proteostasis through complementary nutritional, metabolic, and

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exercise-related mechanisms. Together, these approaches plausibly support cellular maintenance. (**Thymosin α1** and **Thymulin** plausibly support immune surveillance around the FOXO4-DRI senolytic pulses.) [36,37,44,45,73]

Stack Design: Compatibility

Many stack items are in different primary physiologic lanes and their differing primary mechanisms are plausibly compatible, while their downstream effects may overlap and produce complementary, additive effects.

The mitochondrial lane overlap: The mitochondrial and energy supplements have plausibly complementary mechanisms: ALCAR supports mitochondrial substrate entry and acetyl-CoA availability, CoQ10 participates in mitochondrial electron transport and supports ATP production, creatine buffers cellular ATP and stabilizes energy supply, and beta-alanine raises muscle carnosine and buffering capacity, with performance benefits best supported for efforts lasting approximately 1–4 minutes after sustained supplementation, commonly at 4–6 g/day; apigenin is a polyphenol that modulates signaling pathways (AMPK/SIRT, antioxidant responses); EPA and DHA contribute to cellular membrane composition and plausibly support inflammatory regulation and mitochondrial resilience.-Their differing primary roles provide a mechanistic rationale for combined use.

The immune lane overlap: Thymulin and Thymosin α1 act through distinct aspects of thymic and immune regulation. Their combination is intended to provide complementary immunomodulatory signaling and should be coordinated as a shared immune-modulation lane.

The sleep lane overlap: Epitalon, DSIP, and L-Glycine act through distinct but potentially overlapping mechanisms. Their combination is intended to coordinate pineal and circadian signaling, sleep-related neuroendocrine pathways, and glycine-mediated thermoregulatory and neurotransmitter effects.

The annual **Epitalon + Pinealon longevity window** is intentionally run with other peptides paused (except GLP-1s-) to create a clear biologic window and reduce confounding interactions.

Separating **FOXO4-DRI senolytic pulses** from GH-axis on-cycles and repair peptides plausibly reduce overlap between senolytic and anabolic or regenerative signaling.

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Stack Design: Conservative Dosing and Cycles

I use Retatrutide at 1.7 mg twice weekly (3.4 mg/week), with the intention of supporting metabolic health while maintaining my weight.

GH secretagogues are cycled 12 weeks ON and 6 weeks OFF, an empirical schedule that provides a break from GH-axis stimulation and is intended to preserve responsiveness.

I alternate GLOW and GHK-Cu approximately every 3 weeks to limit continuous exposure to TB-500 (TB4).

Overall, many items are intentionally dosed well below community-reported maxima to prioritize safety and tolerability.

Cautions

Informed consent takes on a serious meaning and role in the context of gray-market sourcing, peptides that are not FDA approved, prescriptions that are used in off-label contexts, and protocols that call for near-continuous use.

Specific Cautions

- **Active or recent malignancy or elevated cancer risk**: Tesamorelin and regenerative or potentially angiogenic peptides (GHK-Cu, TB-500 (TB4), BPC-157) warrant particular caution and oncology review before use.
- **Marked baseline IGF-1 elevation or pituitary disease**: GH-axis stimulants warrant caution and should not be initiated until IGF-1 testing and appropriate endocrine evaluation are complete.
- **Uncontrolled diabetes or insulin deficiency**: GLP-1/GIP/glucagon agents and GH-axis stimulants can alter glycemia; unstable diabetes warrants particular caution, and type 1 diabetes requires specialist oversight.
- **Significant hepatic impairment**: Metabolic and hepatic-targeting agents warrant particular caution; decompensated liver disease requires specialist oversight.
- **Severe renal impairment**: Clearance or tolerability of some components may be altered; advanced chronic kidney disease warrants individualized review, including component-specific dosing and monitoring.
- **Pregnancy and breastfeeding**: Peptides and metabolic agents warrant particular caution and generally should be avoided during pregnancy and lactation unless specifically recommended by an appropriate specialist.

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- **Active autoimmune disease or immunosuppressive treatment**: Immune-modulating peptides warrant caution because they may alter immune activity; coordinate their use with the treating specialist.
- **History of pancreatitis or severe gastroparesis**: GLP-1/GIP agents warrant particular caution.
- **Hypotension and syncope risk**: Daily tadalafil and oral minoxidil can lower blood pressure, with potential additive effects when combined. [86,97] Use caution with antihypertensives or alpha-blockers. Nitrates should not be combined with tadalafil. Review potent CYP3A4 inhibitors, which can raise tadalafil exposure.
- **Iron overload**: Avoid supplemental iron with known iron overload or hereditary hemochromatosis. Include iron from the multivitamin, other supplements, and food when assessing total daily intake.[81]
- **Iron and zinc timing**: Take iron and zinc supplements at different times, because taking 25 mg or more of supplemental iron alongside zinc can reduce zinc absorption.[82]

Monitoring

Baseline (before starting or continuing): fasting IGF-1 (use the same lab, assay, and time of day each time), fasting glucose, HbA1c, fasting insulin (HOMA-IR), CMP (AST, ALT, bilirubin, creatinine, electrolytes), serum magnesium, CBC with differential, lipid panel, TSH and free T4, serum copper + ceruloplasmin + zinc (for GHK-Cu), PSA and age-appropriate cancer screening.

Iron status: Check ferritin and transferrin saturation alongside the CBC to assess iron stores and guide continued iron supplementation. Interpret ferritin in the context of inflammation, which can elevate it.[81]

Vitamin B12 status: Check serum vitamin B12 periodically because of vegan dietary intake and metformin use. Consider methylmalonic acid when B12 results are borderline or symptoms suggest deficiency.[85,98]

Repeat relevant laboratory tests periodically, with timing guided by results, symptoms, regimen changes, and clinical judgment. Recheck IGF-1 during a GH-axis break to assess whether it returns toward baseline.

Personal Triggers for Reassessment: These thresholds are personal monitoring triggers for reassessment, interpreted alongside symptoms, repeat testing, and clinical context.

- Persistent IGF-1 above +3 standard deviation scores (SDS) for age: pause GH-axis stimulants and evaluate.

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- Fasting blood glucose increase greater than 20 mg/dL: reassess metabolic agents.
- A1c increase of 0.3 percentage points or more: reassess metabolic agents.
- Aspartate aminotransferase (AST) or alanine aminotransferase (ALT), markers of liver injury, greater than twice the laboratory's upper limit of normal (ULN): repeat testing, investigate other causes, and review retatrutide, tesamorelin, tadalafil, metformin, supplements, and experimental peptides. Consider pausing the component most closely associated in time with the abnormality.
- Serum creatinine increase greater than 30% from baseline: repeat testing, assess hydration and estimated glomerular filtration rate (eGFR), and consider whether creatine supplementation is raising serum creatinine without reducing kidney function. Review metformin and tadalafil because kidney function affects their use, and review retatrutide if vomiting, diarrhea, or dehydration is present.

Biological Mechanisms

Acetyl-L-Carnitine (ALCAR) — an acetylated carnitine derivative and acetyl-group donor. By contributing to the cellular carnitine pool and acetyl-CoA buffering, it plausibly supports fatty-acid utilization, neuronal energetics, and skeletal-muscle mitochondrial function. [65,67]

Apigenin — a naturally occurring flavone with plausible antioxidant and anti-inflammatory activity. Apigenin likely supports restoration of stress-suppressed autophagy. [73] Apigenin likely supports higher hepatic NAD^+ levels through inhibition of CD38. [72] It plausibly influences cellular-stress and senescence-related signaling through pathways involving NF-κB, AMPK, SIRT1, and NAD+ metabolism, with downstream support for mitochondrial, endothelial, and neuronal resilience.

Beta-alanine — a carnosine precursor. Beta-alanine increases muscle carnosine and buffering capacity; performance benefits are best supported for efforts lasting approximately 1–4 minutes after sustained supplementation, commonly at 4–6 g/day. [56] Combined with creatine and mitochondrial support, it plausibly provides complementary support across muscular energy systems.

Beta-sitosterol — a plant sterol that contributes to reduced intestinal cholesterol absorption. LDL-cholesterol lowering is established for plant-sterol intakes of approximately 1.5–3 g/day. [99] It improves urinary symptoms in men with benign prostatic hyperplasia, with plausible anti-inflammatory and membrane-modulating effects in prostate tissue. [38]

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BPC-157 — a synthetic gastric pentadecapeptide. It likely supports tendon, ligament, muscle, and gastrointestinal tissue healing. BPC-157 likely supports restoration of tissue perfusion under ischemic conditions. [104]These effects are plausibly mediated through VEGFR2 and nitric-oxide signaling, angiogenesis, inflammatory modulation, epithelial repair, and stabilization of local microcirculation.

CoQ10 — a mitochondrial electron-transport cofactor that acts as a mobile electron carrier within the inner mitochondrial membrane, accepting electrons from Complexes I and II and transferring them to Complex III. This role supports ATP production. Its redox activity plausibly supports antioxidant defenses, mitochondrial membrane stability, and cardiovascular and metabolic function. A meta-analysis of randomized human trials found that CoQ10 supplementation reduced fatigue across varied study populations. [68]

Creatine monohydrate — supplies creatine for phosphocreatine production and rapid ATP regeneration, supporting cellular energy buffering.

Epitalon — a tetrapeptide acting on pineal and circadian pathways. Epitalon likely supports restoration of age-disrupted melatonin rhythms, with plausible support for telomere maintenance through effects on telomerase activity. [28–30] An annual Epitalon course plausibly influences circadian biology.Longevity effects remain speculative.

GHK-Cu — a copper-binding tripeptide (glycyl-L-histidyl-L-lysine–copper). It likely supports matrix remodeling, wound healing, and dermal and soft-tissue repair through plausible effects on collagen and proteoglycan synthesis, MMP/TIMP balance, inflammatory signaling, and oxidative-stress regulation.

Iron bisglycinate — an oral iron supplement that supplies iron for hemoglobin synthesis, oxygen transport, and iron-dependent cellular processes. In people with iron deficiency, supplementation replenishes iron stores and supports correction of iron-deficiency anemia.[81]

KPV (Lys-Pro-Val) — an anti-inflammatory tripeptide corresponding to the C-terminal sequence of α-melanocyte-stimulating hormone. KPV likely reduces inflammatory-cell recruitment beyond the intestine. [105] It likely reduces intestinal inflammation, with plausible support for mucosal-barrier recovery through effects involving PepT1-mediated cellular uptake, inhibition of NF-κB and MAPK signaling, and reduced pro-inflammatory cytokine production. [17]

L-glycine — a nonessential amino acid that supports collagen synthesis, serves as an inhibitory neurotransmitter in the spinal cord and brainstem, and acts as a key glutathione precursor and one-carbon metabolism substrate, thereby contributing to antioxidant

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defenses and cellular repair. Oral glycine likely promotes peripheral heat loss and lowers core body temperature through NMDA-receptor signaling, plausibly contributing to sleep onset. [100]In small human studies, 3 g of glycine before bedtime improved subjective sleep quality and shortened sleep-onset latency in volunteers reporting unsatisfactory sleep, and improved next-day alertness during partial sleep restriction. [50,87]

Metformin — a biguanide insulin sensitizer. It lowers hepatic glucose output through several complementary mechanisms, including effects on mitochondrial complex I and cellular energy state, AMPK-dependent and AMPK-independent signaling, hepatic redox balance, glucagon signaling, and gut-mediated pathways. These actions reduce gluconeogenesis and support improved glycemic control. Secondary effects on lipid metabolism and inflammation plausibly contribute to broader metabolic benefits.

Minoxidil — a hair-growth treatment that increases hair density in androgenetic alopecia. [42,43] Its effects are plausibly mediated through ATP-sensitive potassium-channel activity, earlier entry of resting follicles into anagen, prolongation of the growth phase, and modulation of VEGF, prostaglandin E2, and Wnt/β-catenin signaling.

β-Nicotinamide mononucleotide (NMN) — an NAD+ precursor that is converted intracellularly to NAD+. Oral NMN increases NAD-related metabolites. NMN likely supports mitochondrial function and NAD^+-dependent DNA repair, with plausible support for redox balance and sirtuin-dependent signaling involved in cellular energy metabolism and resilience. [62,63]

Omega-3 (EPA and DHA) — long-chain omega-3 fatty acids that integrate into cellular and neuronal membranes and support membrane structure and function. Prescription omega-3s at 4 g/day lower elevated triglyceride levels. [88] EPA contributes to bioactive lipid-mediator pathways and plausibly supports inflammation resolution and vascular function. Together, EPA and DHA plausibly support neuronal, endothelial, and metabolic health.

Pinealon (EDR) — a synthetic tripeptide composed of glutamic acid, aspartic acid, and arginine (Glu-Asp-Arg). It likely supports neuronal protection against oxidative stress, with plausible benefits for cognitive resilience. Its effects plausibly involve regulation of antioxidant defenses and cell-survival signaling, with likely support for preservation of neuronal dendritic spines. [33,34,101,102]

PQQ — a redox-active quinone that likely supports mitochondrial biogenesis. [108] PQQ improved memory test scores in human trials using approximately 20 mg/day for 12 weeks. [76,77] It plausibly influences CREB and PGC-1α signaling, mitochondrial number, electron-transport function, and oxidative-stress responses. These mechanisms plausibly

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support neuronal resilience and cellular energy function and complement NMN, ALCAR, and CoQ10 by supporting mitochondrial capacity.

Prostamax (KEDP) — a synthetic tetrapeptide (Lys-Glu-Asp-Pro) from the Khavinson 'bioregulator' family. Some practitioners use it as a prostate-targeted bioregulator with the aim of supporting prostate-tissue regulation, reducing local inflammation, improving microcirculation, and promoting repair. Effects on chromatin structure are plausible based on laboratory findings in human lymphocytes; proposed gene-expression effects and prostate clinical benefits remain speculative. [35]

Retatrutide — a triple agonist at GLP-1, GIP, and glucagon receptors. By adding glucagon-receptor activation to incretin signaling, it likely increases systemic energy expenditure. [103] It plausibly supports fat oxidation through cAMP-mediated lipolysis and mitochondrial substrate switching, while the GLP-1/GIP components support appetite suppression and postprandial insulin dynamics. The combined activity produces substantial reductions in body weight and liver fat.

Semax — a synthetic peptide derived from ACTH fragment 4–10 and delivered intranasally. Semax likely supports neurotrophic signaling, including BDNF-related pathways, with plausible benefits for synaptic plasticity. [25,26] Russian clinical use and regional studies report neuroprotective effects in cerebrovascular settings. Semax likely supports attention and working memory. [27] I use Semax occasionally.

Tadalafil — a selective PDE5 inhibitor that raises intracellular cGMP by preventing its breakdown, thereby potentiating nitric-oxide–mediated vasodilation. It plausibly supports endothelial function, microvascular perfusion, tissue oxygenation, vascular inflammatory regulation, and arterial flexibility. These effects provide a plausible rationale for perfusion-dependent tissue repair and broader metabolic support. [41,86] In a randomized trial of men with BPH and erectile dysfunction, tadalafil 5 mg daily increased circulating eNOS levels and reduced several endothelial-activation and angiogenic biomarkers over 12 weeks. [109]

TB-500 (TB4) — full-length thymosin β4, an actin-binding peptide involved in cytoskeletal regulation. It likely supports tissue repair through effects on cell migration, angiogenesis, and extracellular-matrix remodeling. [14,90,91]

Tesamorelin — a GHRH analogue that binds the pituitary GHRH receptor and increases endogenous pulsatile GH secretion. It likely reduces visceral abdominal fat and liver fat. These effects are plausibly mediated through GH-dependent lipolysis and hepatic lipid metabolism. [9–11]

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Thymosin α1 — a thymic peptide immunomodulator. It likely supports dendritic-cell activation and Th1-mediated antifungal immunity. [107] It plausibly supports T-cell maturation, antigen-presenting-cell function, and immune surveillance by modulating innate and adaptive immune signaling, including Th1 and regulatory pathways, rather than acting as a broad immunosuppressant.

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